IROS 2026 · MOMA.V5 WORKSHOP / ICRA 2026 · CONTACT-RICH ROBOTICS WORKSHOP

Loco-Manipulation with Quadruped Robots

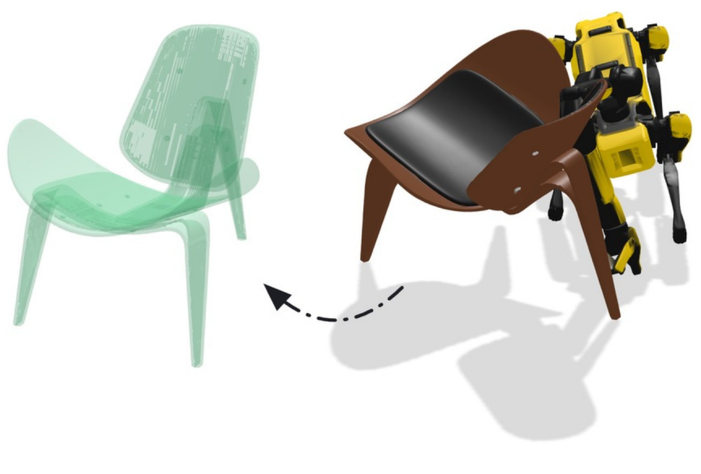
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UNIVERSITY OF MINNESOTA

★ Best Poster Award · MoMA.v5

Spot with an arm A – E

Learn2Push · IROS 2026



(a) Chair · the training object



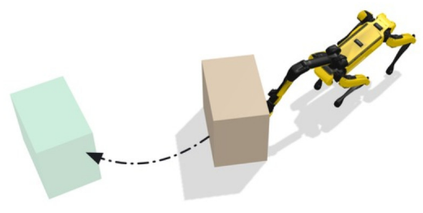
(b) Planter · never seen



(c) Round coffee table · never seen



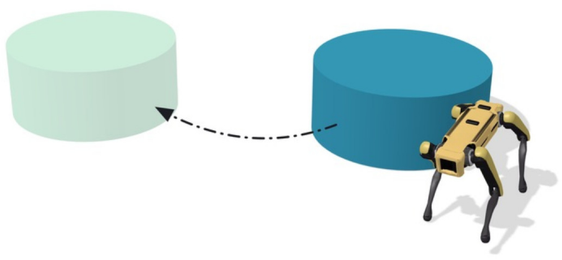
(d) Wire computer cart · never seen



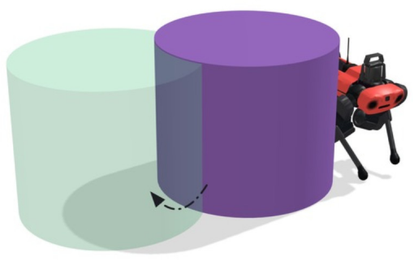
(e) Slippery box · beyond training friction

Legs only F – H · THREE QUADRUPEDS

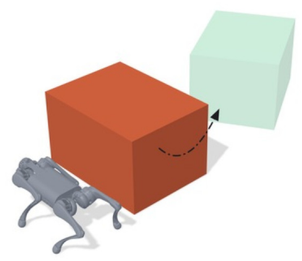
Whole-body pushing · ICRA 2026



(f) Spot · cylinder



(g) ANYmal-C · cylinder



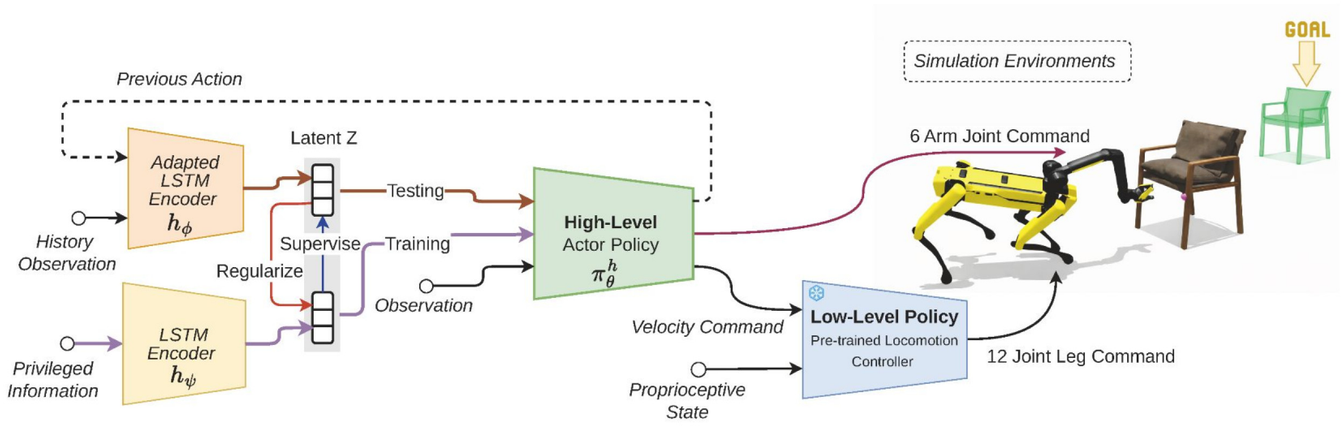
(h) Unitree Go1 · box

What you are seeing. Each panel is a frame from a recorded IsaacLab rollout: the robot pushes the object along the dashed path to its goal pose, shown in green. One two-level policy drives *every* robot. A high-level policy chooses a base velocity, plus six arm-joint targets on Spot with an arm, and a frozen locomotion controller turns that into leg motion.

How it works

TRAINING OBJECTIVE

$$L(\pi_\theta^h, \pi^l, h_\psi) = E_{r\sim p(r\,|\,\pi_\theta^h,\pi^l,h_\psi)}\left[\sum_{t=0}^{T-1}\gamma^t r_t\right] + \lambda\,\|z_\phi - \mathrm{sg}[z_\psi]\|^2 + \|\mathrm{sg}[z_\phi] - z_\psi\|^2$$



Where to touch



With an arm, *where* you push matters. Candidate contact points are sampled on the object's point cloud, filtered, and turned into an intrinsic reward that guides the hand to them.

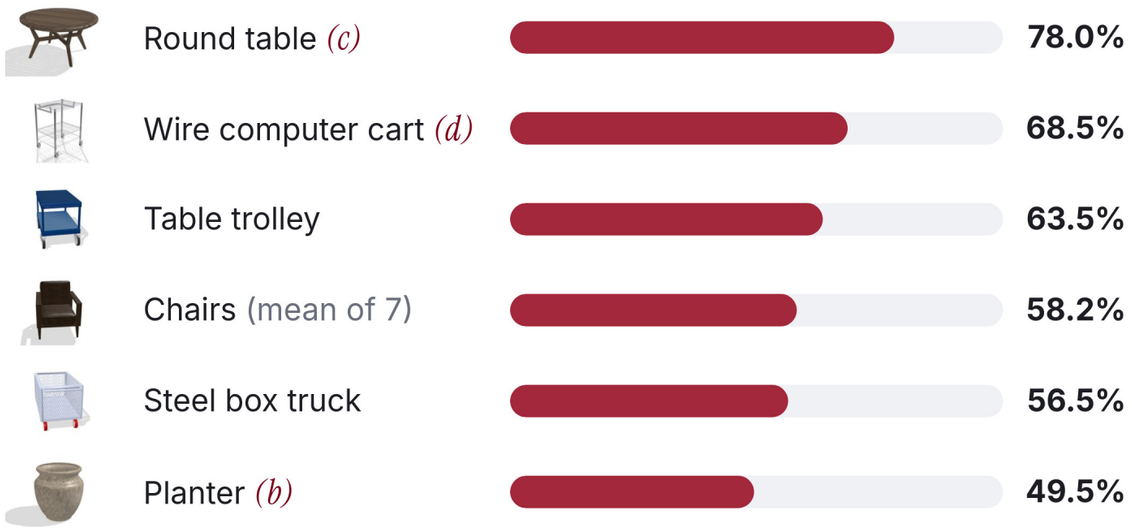
1 **Two levels.** A high-level policy (PPO) decides where to go and how to move the arm; a *frozen* locomotion controller turns that into steps.

2 **Feel the object out.** A teacher sees the object's true properties in simulation; an LSTM student learns the same latent from the robot's *own history*.

3 **Reward in stages.** First *move* it, then *line up* position and heading, then *let it settle*.

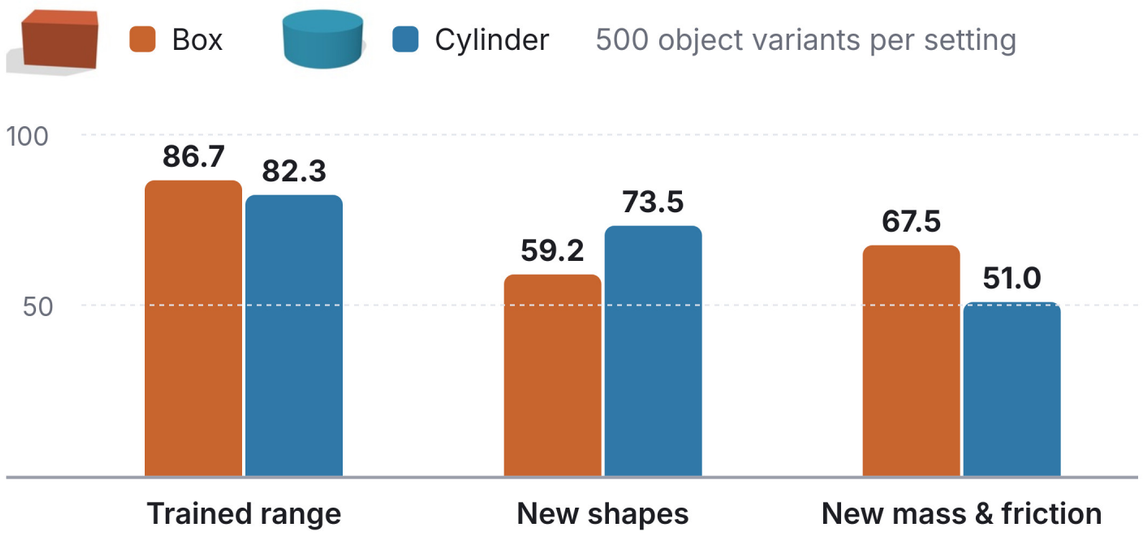
Trained with a single chair. Pushes furniture it has never seen.

Spot with an arm · objects never seen in training



Share of 200 episodes per object ending within 15 cm and 15° of the goal (position only for round table and planter). Chairs: mean of 7 unseen chairs.

Legs only · new shapes and new physics



Share of 500 object variants per setting ending within 15 cm and 15° of the goal: inside the training range, with unseen shapes, and with unseen mass and friction.

Learn2Push
Spot with an arm · project page

Whole-body quadrupedal pushing
three robots · project page

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